Graphical Abstract

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Highlights

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- Research highlight 1
- Research highlight 2

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SingularityNet, , , , ,

Abstract

Abstract text.

Keywords:

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Reinforce

Trajectory or path or episode

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \ldots, s_T) \tag{1}$$

Where:

- s_t: state at time t
- a_t: action
- r_t: reward
- T: episode length

Policy:

$$\pi_\theta(a_t|s_t) \tag{2}$$

Discounted return

todo - infinite vs finite horizon

$$R(\tau) = \sum_{t=0}^{T} \gamma^t r_t \tag{3}$$

This is a discounted return over the whole trajectory. For infinite trajectory we should use $\gamma < 1$ to make it a convergent series, but in practice it's possible to omit this constant e.g. when episodes are short or there is just one final reward etc..

Future discounted returns from time t:

$$G_t = \gamma^t \sum_{k=t}^{T} \gamma^{k-t} r_k \quad \text{or} \quad G_t = \sum_{k=t}^{T} \gamma^k r_k \tag{4}$$

Objective (expected return):

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta(\tau)}[R(\tau)] \tag{5}$$

$$J(\theta) = \int \pi_\theta(\tau) R(\tau) d\tau \tag{6}$$

Policy gradient derivation

Take gradient of the objective:

$$\nabla_\theta J(\theta) = \int \nabla_\theta \pi_\theta(\tau) R(\tau) d\tau \tag{7}$$

Log-derivative trick:

$$\log' f(x) = \frac{f'(x)}{f(x)} \Rightarrow f'(x) = f(x) \log' f(x) \tag{8}$$

Using this identity we get:

$$\nabla_\theta J(\theta) = \int \pi_\theta(\tau) \nabla_\theta \log \pi_\theta(\tau) R(\tau) d\tau \tag{9}$$

Replace integral with expectation:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta(\tau)}[\nabla_\theta \log \pi_\theta(\tau) R(\tau)] \tag{10}$$

In this equation we have for $\pi_\theta(\tau)$:

$$\pi_\theta(\tau) = p(s_0) \prod_{t=0}^{T-1} \pi_\theta(a_t|s_t) p(s_{t+1}|s_t, a_t) \tag{11}$$

if we take log of this product we get sum:

$$\log \pi_\theta(\tau) = \log p(s_0) + \sum_{t=0}^{T-1} \log \pi_\theta(a_t|s_t) + \sum_{t=0}^{T-1} \log p(s_{t+1}|s_t, a_t) \tag{12}$$

Now take gradient with respect to θ, terms not depending on θ drop.

$$\nabla_\theta \log \pi_\theta(\tau) = \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) \tag{13}$$

Plug this expression back to the equation:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta(\tau)} \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) R(\tau) \tag{14}$$

Since past rewards don't depend on current(or future) action at we can replace $R(\tau)$ with G_t.

proof - todo

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta(\tau)} \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) G_t \tag{15}$$

1. Q and Value Functions

Now we can extract Q and value functions from the expection above:

$$Q_{\pi_\theta}(s_t, a_t) = \mathbb{E}_{\tau \sim \pi_\theta(\tau)} \left[\sum_{k=t}^{T} \gamma^{k-t} r_k \mid s_t, a_t \right] \tag{16}$$

$$V_{\pi_\theta}(s_t) = \mathbb{E}_{a \sim \pi_\theta(s)} Q_{\pi_\theta}(a_t, s_t) \tag{17}$$

This is a relative time which allows for bellman expectation form(see below in Useful identities section).

With $G_t = \gamma^t \sum_{k=t}^{T} \gamma^{k-t} r_k$

we can rewrite out gradient to

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta(\tau)} \sum_{t=0}^{T} \gamma^t \nabla_\theta \log \pi_\theta(a_t|s_t) Q^{\pi_\theta}(s_t, a_t) \tag{18}$$

γ^t comes from G_t

in practice we can drop γ^t or use different future discount functions e.g $w(t) = 1/(1 + t/d)$.

γ^t might drop to too small values for long episodes. For example:

$$\gamma = 0.95^{t=150} \approx 1500.00045 \tag{19}$$

vs

$$\frac{1}{1+\frac{t=150}{d=2}} \approx 0.013 \tag{20}$$

State distribution view

We can marginalise distribution of trajectories $\pi_\theta(\tau)$ to $d_t^{\pi_\theta}(s)$ - distribution of states at time t for our policy.

Factor one term for time t inside trajectory summation:

$$\nabla_\theta \log \pi_\theta(a_t|s_t) Q^{\pi_\theta}(s_t, a_t) \tag{21}$$

becomes:

$$\sum_s \sum_a P(s_t = s, a_t = a) \nabla_\theta \log \pi_\theta(a|s) Q^{\pi_\theta}(s, a) \tag{22}$$

Given that

$$P(s_t = s, a_t = a) = P(s_t = s)\pi_\theta(a|s) \tag{23}$$

we get

$$\sum_s P(s_t = s) \sum_a \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) Q^{\pi_\theta}(s, a) \tag{24}$$

This is where the state distribution appears.

Put the factored equation into $\nabla_\theta J(\theta)$:

$$\sum_s \left(\sum_{t=0}^{T} \gamma^t P(s_t = s) \right) \sum_a \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) Q^{\pi_\theta}(s, a) \tag{25}$$

Define the discounted state visitation distribution:

$$d_\gamma^{\pi_\theta}(s) = \sum_{t=0}^{T} \gamma^t P(s_t = s) \tag{26}$$

Note: it is unnormalised.

Proper distribution:

$$d^{\pi_\theta}(s) = (1-\gamma)\sum_{t=0}^{T}\gamma^t d_t^{\pi_\theta}(s) \tag{27}$$

Then

$$\nabla_\theta J(\theta) = \sum_s d^{\pi_\theta}(s)\sum_a \pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s) Q^{\pi_\theta}(s,a) \tag{28}$$

Given that

$$\pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s) = \nabla_\theta \pi_\theta(a|s) \tag{29}$$

we can also write

$$\nabla_\theta J(\theta) = \sum_s d^{\pi_\theta}(s)\sum_a \nabla_\theta \pi_\theta(a|s) Q^{\pi_\theta}(s,a) \tag{30}$$

We can use the formulation above and explicitly sum over actions if possible, but more often in practice we sample actions, so the $\log \pi_\theta$ form comes back:

$$\nabla_\theta J(\theta) = \mathbb{E}_{s\sim d^{\pi_\theta}}\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[\nabla_\theta \log \pi_\theta(a|s) Q^{\pi_\theta}(s,a)] \tag{31}$$

This is called the policy gradient theorem.

Effectively we can replace expectations over trajectories with expectations over state distribution under current policy.

2. Generalized Advantage Estimation (GAE-λ)

todo

Useful identities

Bellman expectation form:

$$V^{\pi_\theta}(s) = \mathbb{E}_{\tau\sim\pi_\theta(\tau)}[r(s,a) + \gamma V^{\pi}(s')] \tag{32}$$

$$Q^{\pi_\theta}(s,a) = \mathbb{E}_{\tau\sim\pi_\theta(\tau)}[r_t + \gamma V^{\pi_\theta}(s_{t+1})|s_t = s, a_t = a] \tag{33}$$

$$Q^{\pi_\theta}(s,a) = \mathbb{E}_{\tau\sim\pi_\theta(\tau)}[r_t + \gamma\mathbb{E}_{a'\sim\pi_\theta(\cdot|s')}[Q^{\pi_\theta}(s',a')]] \tag{34}$$

With relative time G_t:

$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots \tag{35}$$

$$V^{\pi_\theta}(s) = \mathbb{E}_{\pi_\theta}\left[G_t | s_t = s\right] \tag{36}$$